

Mathematical Foundations of Political Methodology

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$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}$$

- Our *test statistic* is:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

- T is distributed t with $N_1 + N_2 - 2$ degrees of freedom.

Equal sample sizes: Democracy and Development

Table 1: Development Levels

Democracy	Autocracy
$N_1 = 6$	$N_2 = 6$
$\bar{X}_1 = 5200$	$\bar{X}_2 = 4500$
$s_1^2 = 400000$	$s_2^2 = 400000$

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$$1 - pt(1.917, 12 - 2) = 0.042$$

Testing Independence

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Table 2: Development Levels and Democracy

	Lowdem	Mediumdem	Highdem	total
Lowdev	40	30	26	96
Mediumdev	5	40	5	50
Highdev	5	20	25	50
total	50	90	56	196

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- $T \sim \chi^2_{(n_{\text{rows}}-1) \times (n_{\text{cols}}-1)}$

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- The degree of freedom is minimum number of independent data that can specify a system.
- In the previous example, we have $(3-1)*(3-1) = 4$ degrees of freedom, and a chi-square test statistic of $\chi^2_4(.95) = 9.48$
- The number of observations minus the number of necessary relations among these observations.

Calculate Marginal Proportions

Table 3: Development Levels and Democracy

	Lowdem	Mediumdem	Highdem	total
Lowdev				96/196
Meddev				50/196
Highdev				50/196
total	50/196	90/196	56/196	196/196

Calculate Expected Proportions under Independence

Table 4: Development Levels and Democracy

	Lowdem	Mediumdem	Highdem	total
Lowdev	$96/196 * 50/196$	$96/196 * 90/196$	$96/196 * 56/196$	$96/196$
Meddev	$50/196 * 50/196$	$50/196 * 90/196$	$50/196 * 56/196$	$50/196$
Highdev	$50/196 * 50/196$	$50/196 * 90/196$	$50/196 * 56/196$	$50/196$
total	$50/196$	$90/196$	$56/196$	$196/196$

Calculate Expected Counts under Independence

Table 5: Development Levels and Democracy

	Lowdem	Mediumdem	Highdem	total
Lowdev	24.9	44	27.4	96.3
Meddev	12.75	23	14.3	50.05
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$$T = \frac{(40 - 24.9)^2}{24.9} + \frac{(30 - 44)^2}{44} + \frac{(26 - 27.4)^2}{27.4}$$

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$$T = \frac{(40 - 24.9)^2}{24.9} + \frac{(30 - 44)^2}{44} + \frac{(26 - 27.4)^2}{27.4} \\ + \frac{(5 - 12.75)^2}{12.75} + \frac{(40 - 23)^2}{23} + \frac{(5 - 14.3)^2}{14.3}$$

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Bootstrapping

- So far we have used the central limit theorem or statistical reasoning to posit distributions.
- Is our data really from the distribution we thought?
- An alternative is to use bootstrapping.

Interpreting Causal Effects

Rubin Causal Model:

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$$\text{ATE} = E[Y(1) - Y(0)]$$

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- **Regression is just one method**

The Problem of Selection

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- Requires no additional assumptions

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$$\begin{aligned}ATE &= \{E[Y(1)|T = 1]Pr(T = 1) + E[Y(1)|T = 0]Pr(T = 0)\} \\&\quad - \{E[Y(0)|T = 0]Pr(T = 0) + E[Y(0)|T = 1]Pr(T = 1)\} \\&= E[Y(1)|T = 1]\pi + E[Y(1)|T = 0](1 - \pi) \\&\quad - E[Y(0)|T = 0](1 - \pi) - E[Y(0)|T = 1]\pi\end{aligned}$$

Manski Bound

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$$[\text{Average}(\text{Treat})\pi - (1 - \pi)\text{Average}(\text{Control}) - \pi,$$
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- $\pi = 0.06$
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- Further assumptions can narrow bounds
- **Law of Decreasing Credibility:** the credibility of inference decreases with the strength of the assumptions maintained

Causal Inference Via Stratification

Selection on observables:

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- Suppose that treatment is systematically related to outcomes (not ignorable)

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Proposition

Suppose there exists S such that 1) and 2) hold. Then, we can obtain unbiased estimates for

- 1) $E[ATE|S = 1] \equiv ATE$, Given $S = 1$
- 2) $E[ATE|S = 0] \equiv ATE$, Given $S = 0$
- 3) ATE

Idea:

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- 3) Average Treatment Effect

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 - Calculate Average(Treatment) - Average(Control) within each strata, $E[ATE|S]$

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 - Average(Treatment) - Average(Control) , for all units with $S = 0$
- 3) Average Treatment Effect
 - Calculate Average(Treatment) - Average(Control) within each strata, $E[ATE|S]$
 - $ATE = \sum_{s=0}^1 E[ATE|S] \times Pr(S)$

Proof

$$E[Y(1)|S = 1, D = 1] - E[Y(0)|S = 1, D = 0]$$

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$$\begin{aligned} & E[Y(1)|S = 1, D = 1] - E[Y(0)|S = 1, D = 0] \\ &= \{ \Pr(D = 1|S = 1)E[Y(1)|S = 1, D = 1] \\ &\quad + \Pr(D = 0|S = 1)E[Y(1)|S = 1, D = 1] \} \\ &\quad - \{ \Pr(D = 1|S = 1)E[Y(0)|S = 1, D = 0] \\ &\quad + \Pr(D = 0|S = 1)E[Y(0)|S = 1, D = 0] \} \end{aligned}$$

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Causal Inference via Stratification: Example

We are interested in the causal effect of **incumbency** on reelection.

- $T = 1$, Incumbent
- $T = 0$, Challenger

$Y_i(T)$ result of election.

Incumbency obviously not assigned at random.

But suppose we have a dichotomous measure of candidate quality

- $S = 1$, High quality
- $S = 0$, Low quality

And that incumbency is as good as random, given S .

We're interested in obtaining

$$ATE = E[Y(1) - Y(0)]$$

What if we don't condition on candidate quality?

Joint Distribution of Treat and Strata

	$T = 1$	$T = 0$	$P(S)$
$S = 1$	0.4	0.2	0.6
$S = 0$	0.1	0.3	0.4
$P(T)$	0.5	0.5	

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	Control	Treat	$ATE S$
$S = 1$	$E[Y(0) S = 1] = 0.5$	$E[Y(1) S = 1] = 0.7$	0.2
$S = 0$	$E[Y(0) S = 0] = 0.38$	$E[Y(1) S = 0] = 0.4$	0.02

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$$E[Y(0)|T = 0] = 0.5 \times \frac{2}{5} + 0.3 \times \frac{3}{5} = 0.38$$

$$E[Y(1)|T = 1] - E[Y(0)|T = 0] = 0.64 - 0.38$$

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 E[Y(1)|T = 1] - E[Y(0)|T = 0] &= 0.64 - 0.38 \\
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Why?: those with bigger treatment effects more likely to select into treatment

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 &\quad + 0.4 \times P(S = 0|T = 1) \\
 &= 0.7 \times \frac{4}{5} + 0.4 \times \frac{1}{5} = 0.64
 \end{aligned}$$

$$\begin{aligned}
 E[Y(0)|T = 0] &= 0.5 \times \frac{2}{5} + 0.3 \times \frac{3}{5} = 0.38
 \end{aligned}$$

$$\begin{aligned}
 E[Y(1)|T = 1] - E[Y(0)|T = 0] &= 0.64 - 0.38 \\
 &= 0.26
 \end{aligned}$$

Why?: those with bigger treatment effects more likely to select into treatment

Confound effect of T with differences across S

With Stratification

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Exact stratification is **Exact Matching**

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Exact Matching, Basic idea:

- Identify all **characteristics** (covariates) that affect both outcome and treatment assignment
- For all treated units, identify **control** unit with same characteristics
- Exact match: units in the same strata

What do we do?

Most strata are empty, or only a few observations

- Bias-Variance tradeoff
 - Bias: assume same casual effect across strata
 - Variance: assume different causal effect across strata
- Modeling $E[Y|X_1, X_2, X_3, \dots, X_K]$
 - Nonparametric (loess): different curse of dimensionality problem
 - High dimensional space is sparse, hard to borrow across

Solution: specify a model of how covariates relate to treatment

Linear Regression

Consider one continuous covariate S_i and a continuous dependent variable Y_i .

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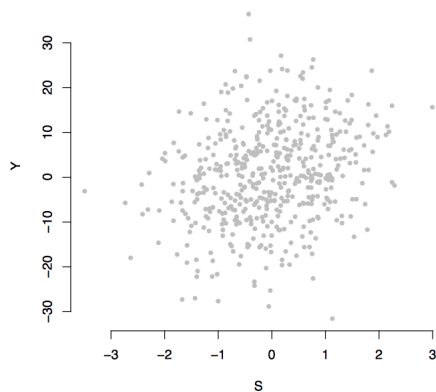
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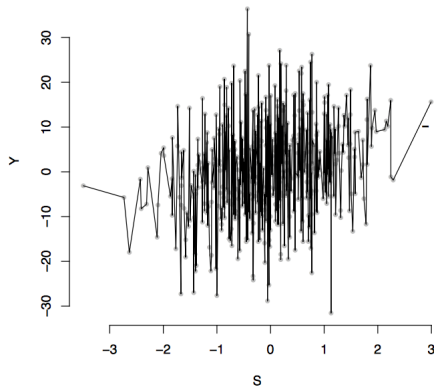
What does this say?

- **Stratifying** (conditioning): examining means of Y given values of S
- **Borrowing** information across bins:
 - **Assuming** that means have a **global** and **linear** movement with S , β_1

Linear Regression

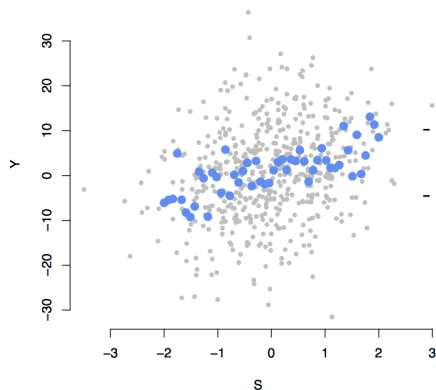


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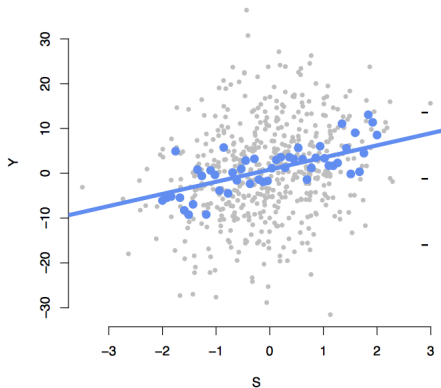
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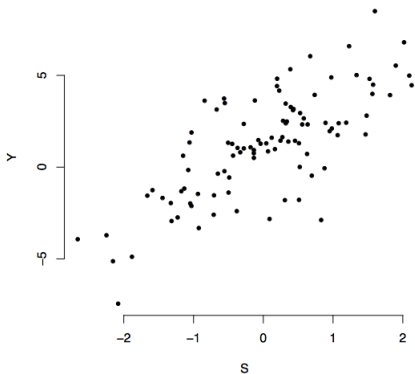
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Graphically:



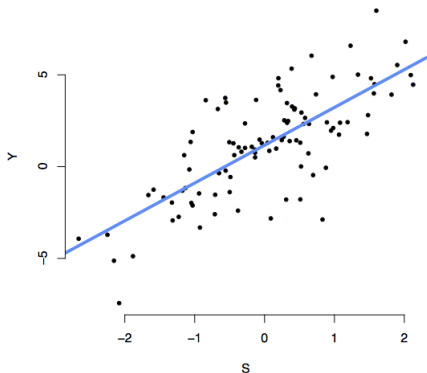
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